Nikhil Dhawan

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Education

Cornell University — Ithaca, NY

Bachelor of Science, Computer Science, *cum laude* — *Aug. 2015 - May 2019*

- Hunter R. Rawlings III Presidential Research Scholar
- Minors in Electrical & Computer Engineering and Business
- Major GPA: 3.733, Cumulative GPA: 3.573

Professional Experience

Google DeepMind — New York, NY

Senior Research Engineer — *Nov. 2021 - Present*

- Core contributor to the Gemini post-training code area
- Leading modeling projects to improve Gemini's coding and SWE agent capabilities through the development of multistep reinforcement learning recipes, data quality pipelines, and well-calibrated frontier coding evaluation sets
- Developed novel agentic-RL research recipes that improved Gemini's performance on SoTA coding evals by XX points as a lead on a Gemini coding sprint before the 2.5 Pro launch
- In a several month long research project, developed data pipelines and RL environments to mine executable coding prompts from a high-quality held-out dataset. These datasets and environments resulted in significant improvements to Gemini's general SWE agent capabilities
- Leveraging the aforementioned datasets, led the development of a multistep SWE-style coding eval that is currently used across Gemini post-training to measure SWE performance
- Eval lead on sprint for improving quality of Google's internal coding agent. Developed the first Google-SWE eval enabling Google's coding agent team to find gaps and prioritize new agentic features
- Formerly, core contributor to code transform model training for Google. Built early high-quality SFT training datasets to teach Gemini how to write multifile code patches
- Promoted L4 $\rightarrow$ L5 in one year due to high performance and impact

Microsoft — Redmond, WA

Software Engineer & Data Scientist — *Aug. 2019 - Nov. 2021*

- Engineered ML models, data systems, and pipelines, conducted efficiency research for the Azure Compute fleet, and reported to the Senior Director of Economics in Azure
- Created and maintained automatic data ingestion pipelines using pySpark, Databricks, Azure Data Factory, and an internal ETL tool
- Engineered statistical models leveraging large datasets of Azure usage data to scientifically determine marginal costs, customer price sensitivity, and optimal prices for several Azure products
- Developed a deep learning model with TensorFlow to predict the probability an Azure customer will experience an allocation failure
- Pricing changes as a result of my work led to > $100mm in additional yearly revenue and reduction in COGS for Microsoft
- Mentored a summer intern: supervised project, taught Python, pandas, and Azure Cloud suite

Human Robot Collaboration Group — Ithaca, NY

Undergraduate Researcher — *Feb. 2016 - May 2019*

- Implemented real-time computer vision in Java to recognize objects with fiducial markers
- Architected a low-latency multi-threaded system with parallelized back-end computations and a mixed-reality UI
- Co-authored best paper at the 2018 DCC Conference, paper at the 2020 IVA Conference, and paper at the 2021 DIS Conference

The Blackstone Group — New York, NY

Software Engineering Intern — *May 2018 - Aug. 2018*

- Rearchitected data-retrieval pipeline from an HTTP request model to an asynchronous publisher-subscriber model
- Eliminated timeouts from large web requests in a C# and .NET system using SignalR web sockets
- Reduced cache read time by 80% using GZip compression
- Implemented lazy-loading of data requests on the UI in JavaScript for improved page-load times

Leadership & Teaching

Cornell Venture Capital Ithaca, NY
Senior Project Manager *Feb. 2017 - May 2019*

- Performed market analysis and due diligence, constructed slide decks, and presented research to associates and partners at leading VC firms such as Google Ventures, Citi Ventures, Bessemer Venture Partners, and Envision Ventures
- Won 2nd pl. at the National Venture Capital Investment Competition

CS 4780 - Machine Learning Ithaca, NY
Teaching Assistant *Jan. 2018 - Dec. 2018*

- Previously TA for CS 4410 - Operating Systems
- Held office hours to instruct and help students, created assignments and homework, and graded student projects and examinations

Publications & Patents

- G Comanici, E Bieber, M Schaekermann, I Pasupat, N Sachdeva, I Dhillon, **N Dhawan**, . . . **Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities.** arXiv preprint arXiv:2507.06261, 2025. https://arxiv.org/abs/2507.06261
- FFE Kohle, JA Hinckley, **N Dhawan**, UB Wiesner. **Sulfur- or heavy atom-containing nanoparticles, methods of making same, and uses thereof.** US Patent 12,397,067, 2025. https://patents.google.com/patent/US12397067B2/en
- MV Law, Z Li, A Rajesh, **N Dhawan**, A Kwatra, G Hoffman. **Hammers for Robots: Designing Tools for Reinforcement Learning Agents.** Proceedings of the 2021 ACM Designing Interactive Systems Conference, pages 1638-1653, 2021. https://dl.acm.org/doi/pdf/10.1145/3461778.3462029
- MV Law, A Kwatra, **N Dhawan**, M Einhorn, A Rajesh, G Hoffman. **Design intention inference for virtual co-design agents.** Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents, 2020. https://dl.acm.org/doi/pdf/10.1145/3383652.3423861
- FFE Kohle, JA Hinckley, S Li, **N Dhawan**, WP Katt, JA Erstling, . . . **Amorphous quantum nanomaterials.** Advanced Materials 31(5), 1806993, 2019. https://advanced.onlinelibrary.wiley.com/doi/am-pdf/10.1002/adma.201806993
- MV Law, **N Dhawan**, H Bang, SY Yoon, D Selva, G Hoffman. **Side-by-side human–computer design using a tangible user interface.** International Conference on Design Computing and Cognition, pages 155-173, 2018. https://hrc2.io/assets/pdfs/papers/LawDhawanBangSelvaYoonHoffman18.pdf

Awards

- **Best Paper Award**: Awarded to the best peer-reviewed paper at the 2018 Design Computing & Cognition Conference
- **CS Departmental Honors**: Awarded to CS undergraduates who maintain a high GPA while pursuing academic research and graduate-level coursework
- **Hunter R. Rawlings III Presidential Research Scholar**: Research grant awarded to less than 1% of the incoming Cornell class for demonstrated excellence in scientific research
- **Salutatorian**: Spackenkill High School Class of 2015
- **Intel ISEF 4$^{\text{th}}$ Place**: Awarded to the 4th place winner in the Material Science category at the 2015 Intel International Science and Engineering Fair

Skills

Languages: Python, C++, Java, C#, C, SQL, TypeScript, OCaml, Matlab

Tools/Frameworks: TensorFlow, Azure, pySpark, Databricks, React, ROS, git, .NET, SignalR, pandas, numpy, LaTeX